

Chapter 17 - JPA Complete Guide

1. What is JPA?

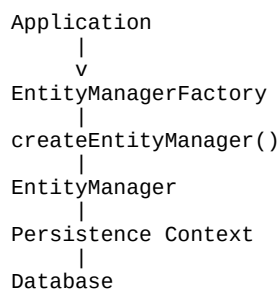
JPA (Jakarta Persistence API) is a specification that defines a standard API for object-relational mapping in Java. It is not an ORM framework by itself.

2. JPA vs Hibernate

Hibernate is one implementation of the JPA specification. Applications written against JPA APIs can switch to another JPA provider with fewer code changes.

JPA	Hibernate
Specification	Framework / JPA Implementation
Defines interfaces	Implements JPA interfaces
Portable API	Provides extra features
EntityManager	Session

3. JPA Architecture



4. EntityManagerFactory

EntityManagerFactory is heavyweight, thread-safe and usually created once for each persistence unit.

5. EntityManager

EntityManager is the primary JPA interface used to persist, retrieve, update and remove entities. It manages the Persistence Context.

6. Persistence Unit

A persistence unit defines entity classes and database configuration. In standard JPA it is configured in persistence.xml. Frameworks such as Spring Boot may configure this programmatically.

7. Persistence Context

The Persistence Context contains managed entity instances. It tracks changes, maintains identity and performs dirty checking.

8. Common EntityManager Methods

```
persist(entity)
find(Employee.class, id)
getReference(Employee.class, id)
merge(entity)
remove(entity)
flush()
clear()
detach(entity)
refresh(entity)
```

9. Entity Lifecycle

Entities transition through Transient, Managed (Persistent), Detached and Removed states during their lifecycle.

10. Transactions

Database modifications should be performed inside transactions. In enterprise applications, transaction management is often provided by Spring using `@Transactional`.

11. Best Practices

- Prefer JPA APIs in new applications.
- Use EntityManager instead of Hibernate Session unless Hibernate-specific features are required.
- Keep Persistence Contexts short-lived.
- Use LAZY fetching where appropriate.
- Use optimistic locking with `@Version` when needed.

12. Interview Questions

1. What is JPA? 2. Is JPA a framework? 3. Difference between EntityManager and Session? 4. What is EntityManagerFactory? 5. What is a Persistence Unit? 6. JPA vs Hibernate? 7. What is the Persistence Context?